CloudFormation Change Sets and Stack Names

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Overview

A change set is a reviewable preview of what CloudFormation would change in a stack, and every change set belongs to exactly one stack. The stack name is how you say which one.

The usual flow is create, review, then execute. Nothing changes in the stack until you execute. This page explains where the stack name is used, and includes a small sample template you can run to see each step.

How the stack name is used

The stack name identifies the stack, and it is unique within one AWS account and region. Every change set operation uses it.

Step	Command	Stack name
Create	`create-change-set` or `deploy --no-execute-changeset`	Required
Review	`describe-change-set`	Required, or use the change set ARN
Execute	`execute-change-set`	Required, or use the change set ARN
List	`list-change-sets`	Required
Delete	`delete-change-set`	Required, or use the change set ARN

When creating, the stack name decides what kind of change set you get. If the stack exists, the change set is an update to it. If it does not, CloudFormation creates an empty stack in the `REVIEW_IN_PROGRESS` state and the change set describes creating everything. A typo in the name therefore produces a change set for a new stack and no error, so check that the review shows modifications and not a list of additions.

A change set name only has to be unique within its stack, so two stacks can each have a change set called `my-update`. A stack can hold several pending change sets. Executing one changes the stack, so the others are then based on an old version of it and usually fail if you try to run them.

Stack names are unique per region, not per account. `leap-team1` in us-east-1 and `leap-team1` in another region are different stacks, so always pass the region too. A deleted stack can no longer be named, and its name can be reused by a new stack. Its stack ID (the stack's ARN) can still be used to look at it.

Sample template

Save this as `demo.yaml`. It creates one empty S3 bucket and one SSM parameter, both named from the stack name, so you can see the stack name appear inside the resources. Use a lowercase stack name, because bucket names must be lowercase. The YAML was checked for syntax but not deployed.

```
AWSTemplateFormatVersion: '2010-09-09'
Description: Change set demo. Resource names are built from the stack name.
Parameters:
  Greeting:
    Type: String
    Default: hello
Resources:
  DemoBucket:
    Type: AWS::S3::Bucket
    Properties:
      BucketName: !Sub '${AWS::StackName}-${AWS::AccountId}-${AWS::Region}'
      Tags:
        - {Key: Stack, Value: !Ref 'AWS::StackName'}
  DemoParameter:
    Type: AWS::SSM::Parameter
    Properties:
      Name: !Sub '/${AWS::StackName}/greeting'
      Type: String
      Value: !Ref Greeting
Outputs:
  BucketName: {Value: !Ref DemoBucket}
  ParameterName: {Value: !Ref DemoParameter}
```

Demo walkthrough

1. **Create the stack.** The first run has nothing to compare against, so it creates everything.

```
aws cloudformation deploy --stack-name change-demo --template-file demo.yaml
```

2. **Propose a small change without applying it.**

```
aws cloudformation deploy --stack-name change-demo --template-file demo.yaml
\
  --parameter-overrides Greeting=goodbye --no-execute-changeset
aws cloudformation list-change-sets --stack-name change-demo
aws cloudformation describe-change-set --stack-name change-demo --change-set-
name <name from the list>
```

The review shows `DemoParameter` as Modify with `Replacement: False`. Nothing has changed yet.

3. **Apply it.**

```
aws cloudformation execute-change-set --stack-name change-demo --change-set-
name <same name>
```

4. **See a replacement.** Change `BucketName` in the file, for example by adding `-v2` at the end, and repeat step 2. The bucket now shows `Replacement: True`: CloudFormation would create a new bucket and delete the old one.

5. **See what a wrong stack name does.** Run step 2 with `--stack-name change-demo2`. Every resource shows as Add, and a new stack is left in `REVIEW_IN_PROGRESS`.

6. **Clean up.** Both buckets are empty, so the deletes will not fail.

```
aws cloudformation delete-stack --stack-name change-demo
aws cloudformation delete-stack --stack-name change-demo2
```

To dump the latest changeset and review:

```
aws cloudformation list-change-sets \
 --region us-east-1 \
 --stack-name <team-stack-name> \
 --query 'Summaries[*].[ChangeSetName,Status,ExecutionStatus,CreationTime]' \
 --output table
```

Ex: awscli-cloudformation-package-deploy-1792438618

Describe the list of changes:

```
aws cloudformation describe-change-set \
 --region us-east-1 \
 —stack-name<team-stack-name> \
 --change-set-name  awscli-cloudformation-package-deploy-1792438618   \
 --output json > team4-latest-changeset.json
```

Note: When applying change set:

ECS task definition gets a revision

New ALB target group creates a new target group

To Execute the Changes via ChangeSet:

```
aws cloudformation execute-change-set \
 --region us-east-1 \
 --stack-name <team-stack-name> \
 --change-set-name awscli-cloudformation-package-deploy-1792438618
```

Monitor the Change set being applied:

```
aws cloudformation wait stack-update-complete \
 --region us-east-1 \
 --stack-name <team-stack-name>
```